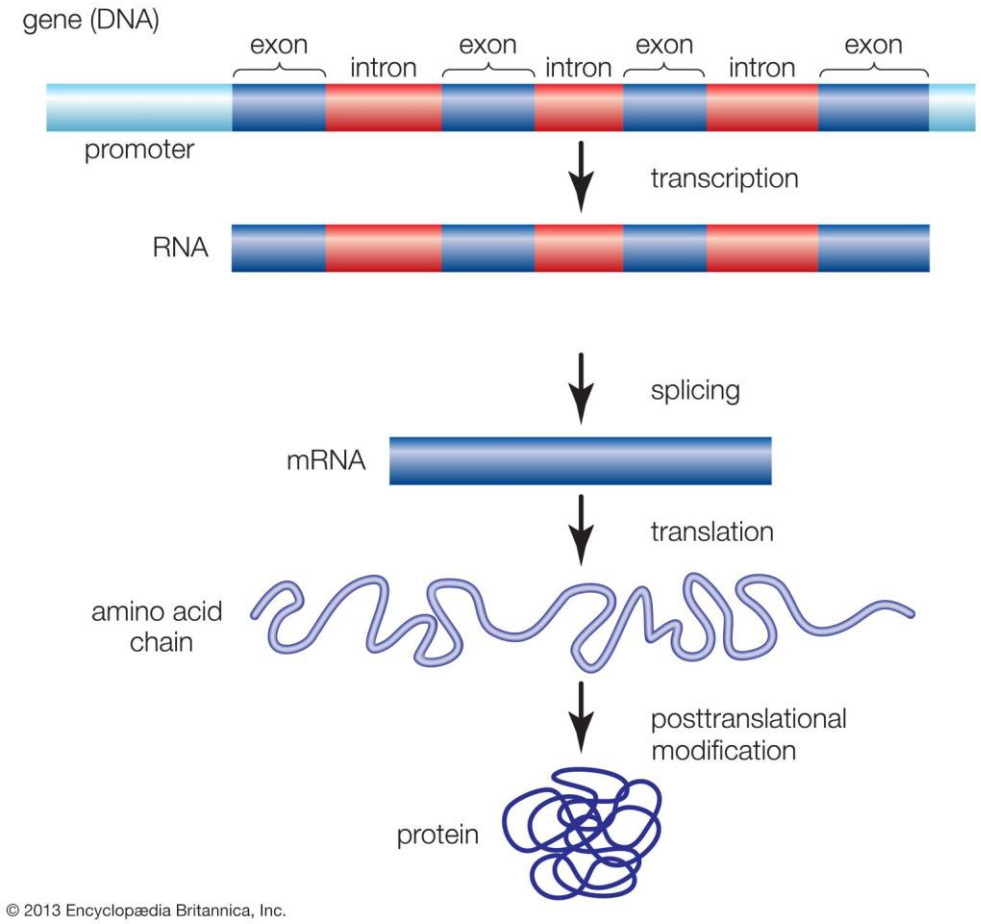
Systems Biology

Introduction to Systems Biology

Molecular Biology

In molecular biology, gene structure and function is studied at the molecular level.



Molecular Biology Experiment

Stem cell **Blood cell**

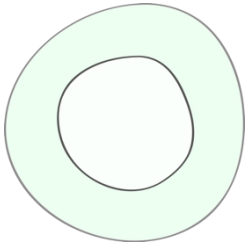


Hypothesis : Gene *Matr3* is required for proper blood cell differentiation.

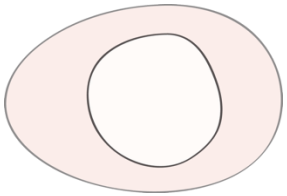
How can we test this?

Molecular Biology Experiment

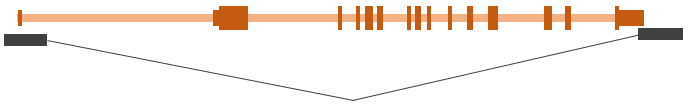
Stem cell



Blood cell

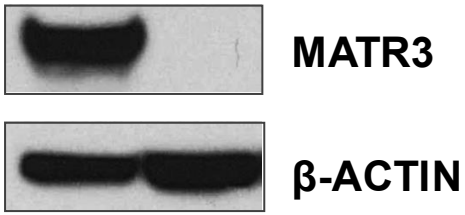


Matr3



Δ

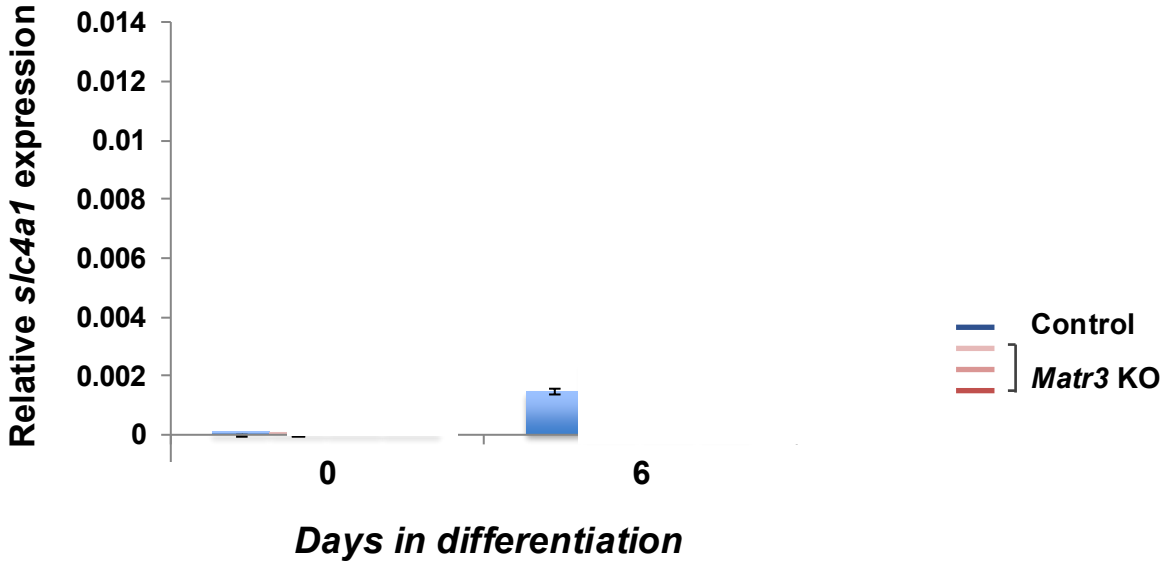
Control *Matr3* KO



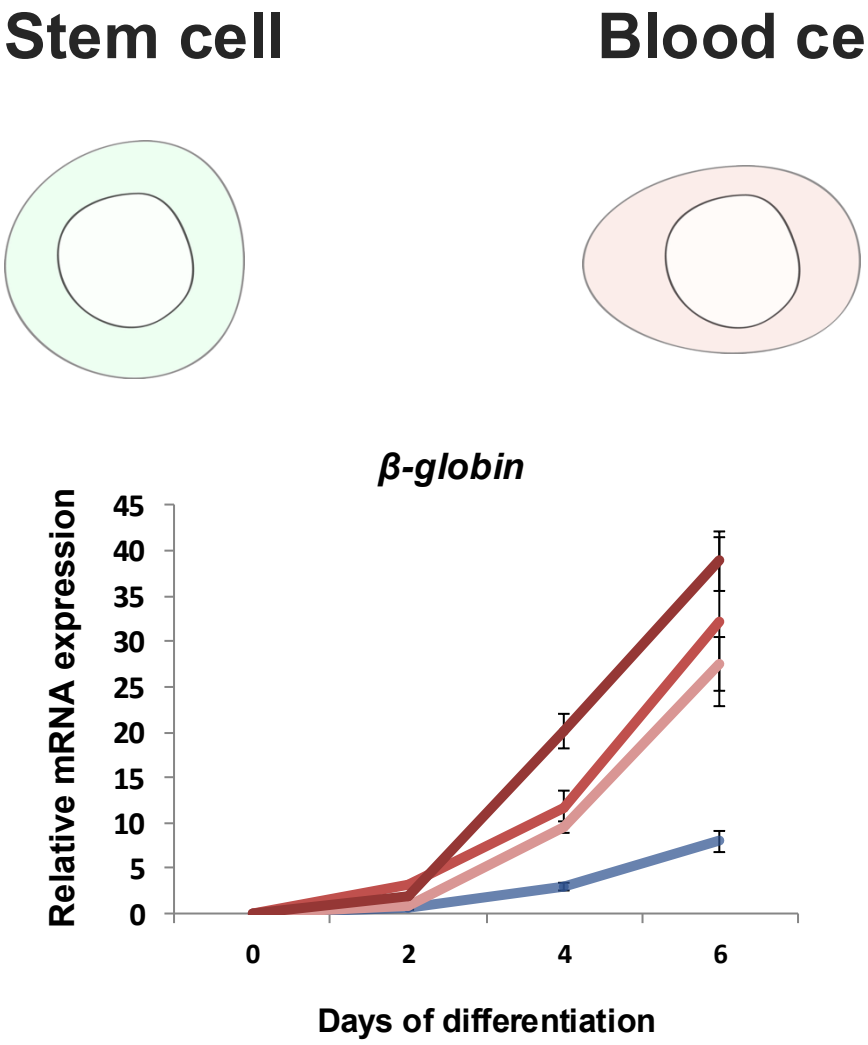
Molecular Biology Experiment

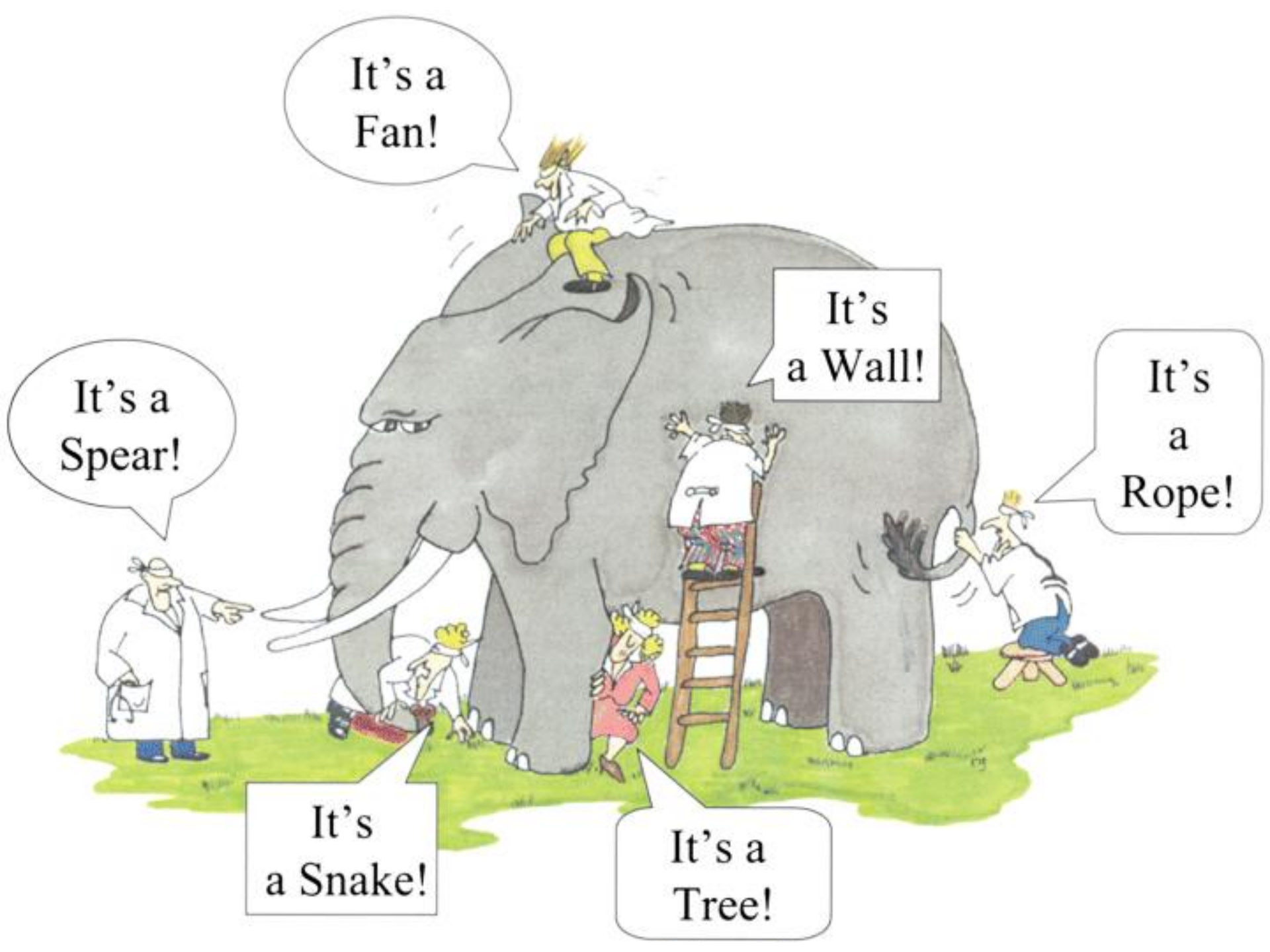
Stem cell **Blood cell**





Molecular Biology Experiment

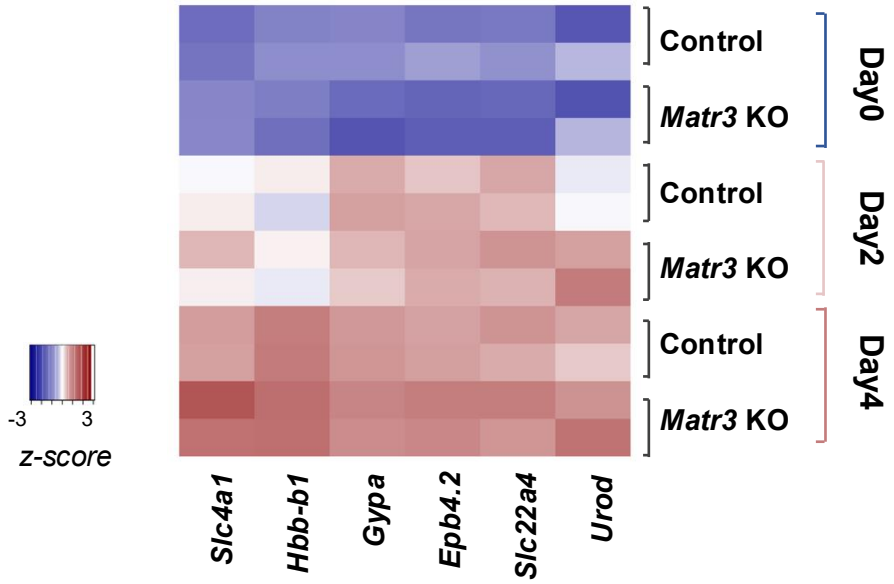


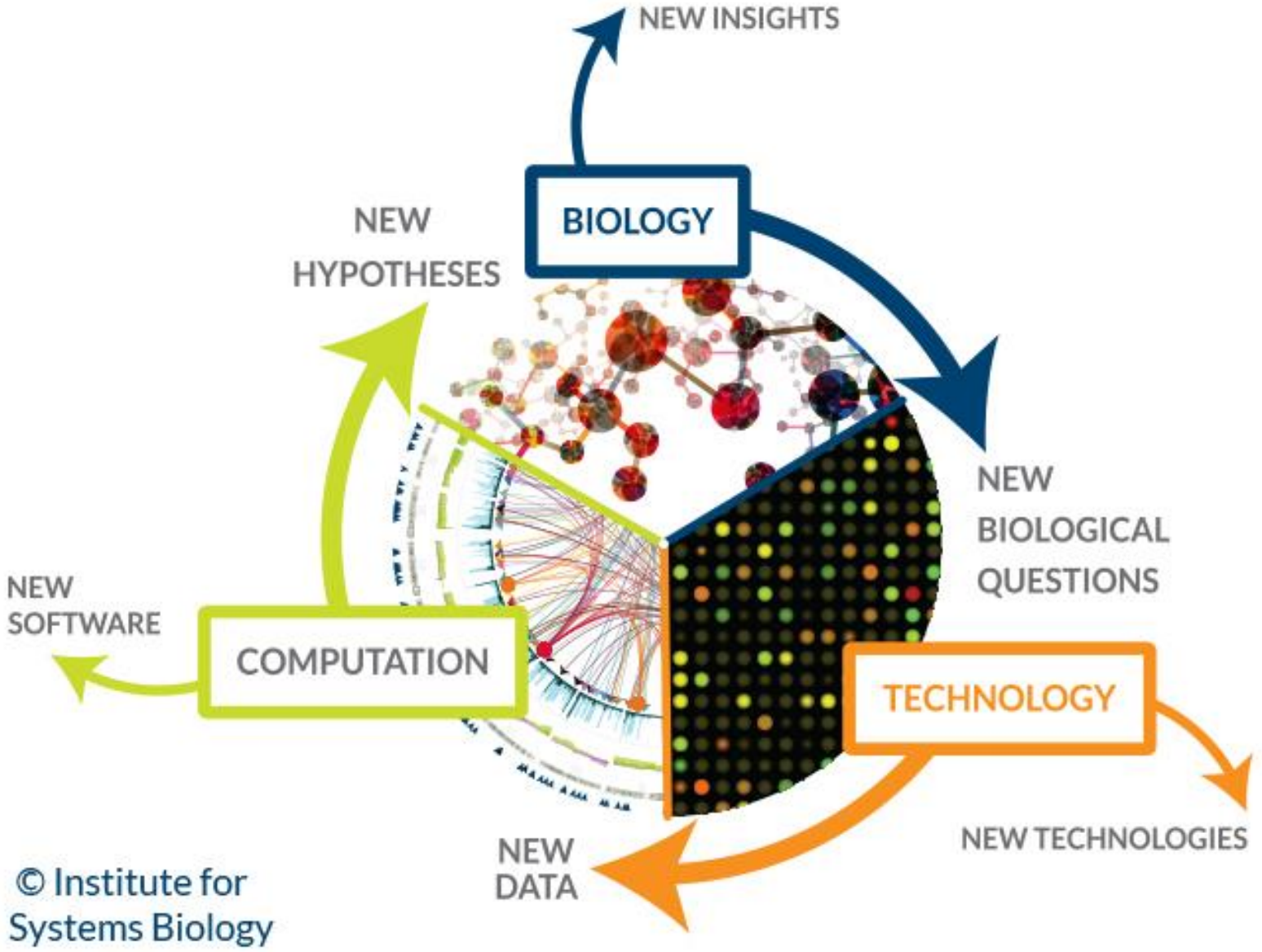

It's a Fan!
It's a Spear!
It's a Wall!
It's a Rope!
It's a Snake!
It's a Tree!

System/Genome Level Experiment

Stem cell **Blood cell**






NEW INSIGHTS
BIOLOGY
NEW HYPOTHESES
NEW BIOLOGICAL QUESTIONS
NEW SOFTWARE
COMPUTATION
TECHNOLOGY
NEW TECHNOLOGIES
NEW DATA
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Midterm project

Oral Presentation Guidelines

1. **Paper Selection**
   - Choose a research paper related to Systems Biology.
   - The paper must be an original research article containing actual experimental/analytical data (not a review paper).
   - The paper must include system-level, data-driven quantitative results — such as multi-omics data or network/modeling-based analysis.

Midterm project

Oral Presentation Guidelines

2. Presentation Format (Choose One)

- Select one of the following two formats for your presentation.
  - [A] Critical Redesign: Choose a specific experiment or analysis from the paper and explain how you would have designed it differently (with reasoning).
  - [B] Analysis Explanation: Choose the analysis or result you found most interesting and explain the method, result, and significance in your own words (including why you found it interesting).

- Required elements (regardless of which format you choose):
  - Brief background and purpose of the study
  - How this connects to or could extend your own research interests

- Presentation time: 5 minutes

Midterm project

Oral Presentation Guidelines

3. Presentation Style

- Present without a script or notes — use only your slides as a guide.
- Slides should favor figures, data, and key terms over complete sentences (avoid reading full sentences off the slide).
- Slides must be submitted by 5 PM the day before your presentation.

4. Question Duty (Rotating)

- Students scheduled to present the following class must ask at least one question during that day's session.

5. Bonus Points

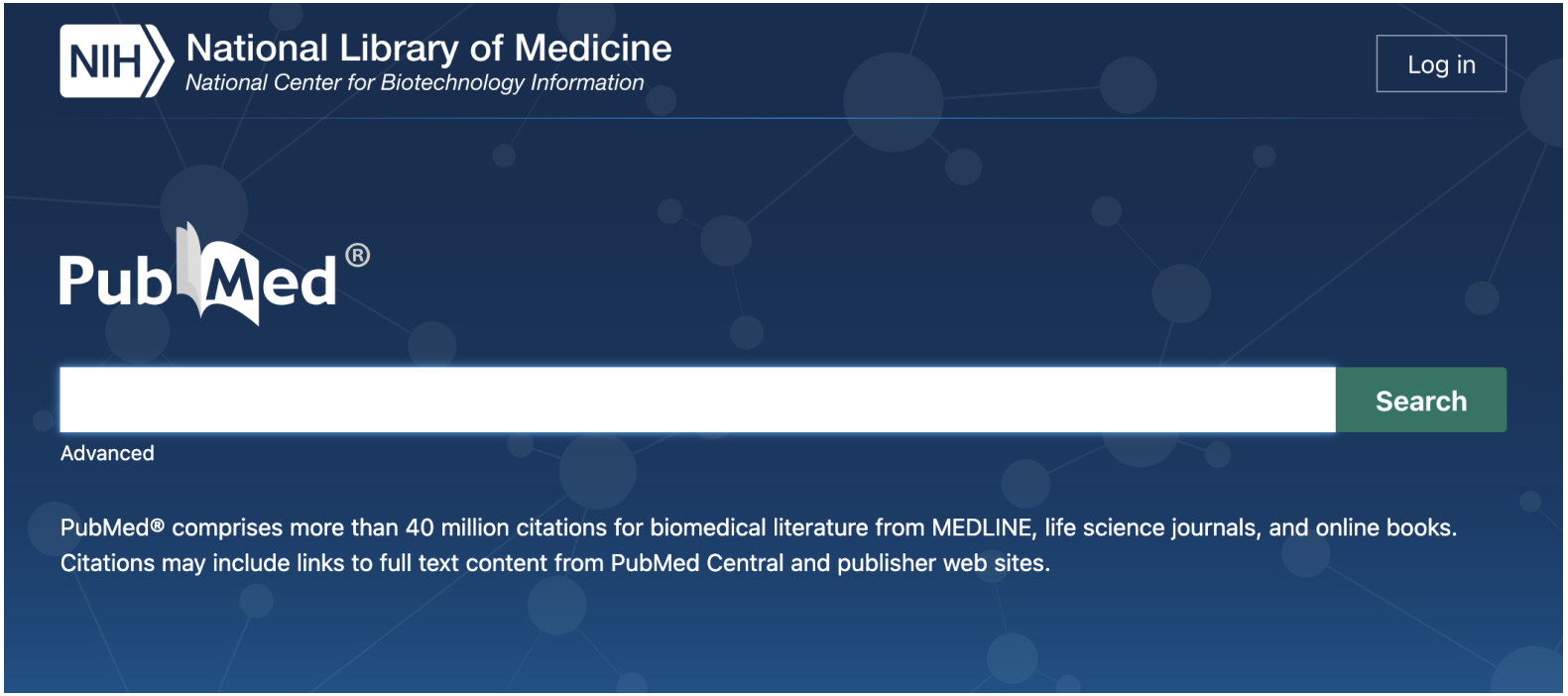
- Bonus points will be given for good questions and good answers.
- Evaluation criteria: whether the question reflects genuine understanding of the paper/presentation, and whether the answer is logical.

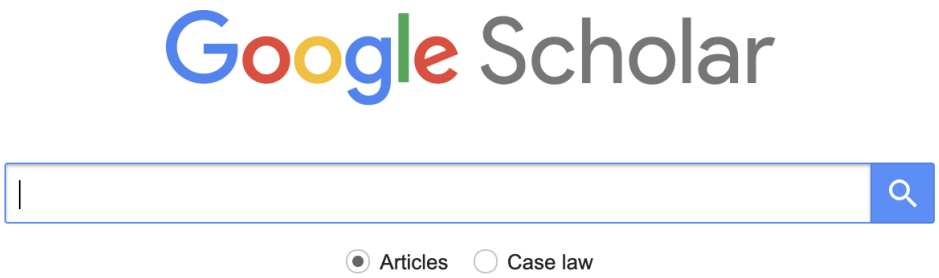
Midterm project

Evaluation Rubric

Criteria	Details	Weight
Understanding of the Paper	Accurate summary of research background/purpose; paper meets requirements (original research, multi-omics/network-based)	20%
Depth of Analysis (A or B)	[A] Redesign logic is sound and specific / [B] Method-result-significance explanation is clear and reformulated in the student's own words	25%
Connection to Own Research	Clear and specific link to how this connects to or extends the student's own research interests	15%
Presentation Skills	Delivered without script/notes; slides use figures/keywords rather than full sentences; adheres to time limit	15%
Q&A Response	Understanding demonstrated and logical reasoning shown in impromptu answers	15%
Question Contribution (for the following week's presenters)	Quality of questions (reflects genuine understanding)	10%

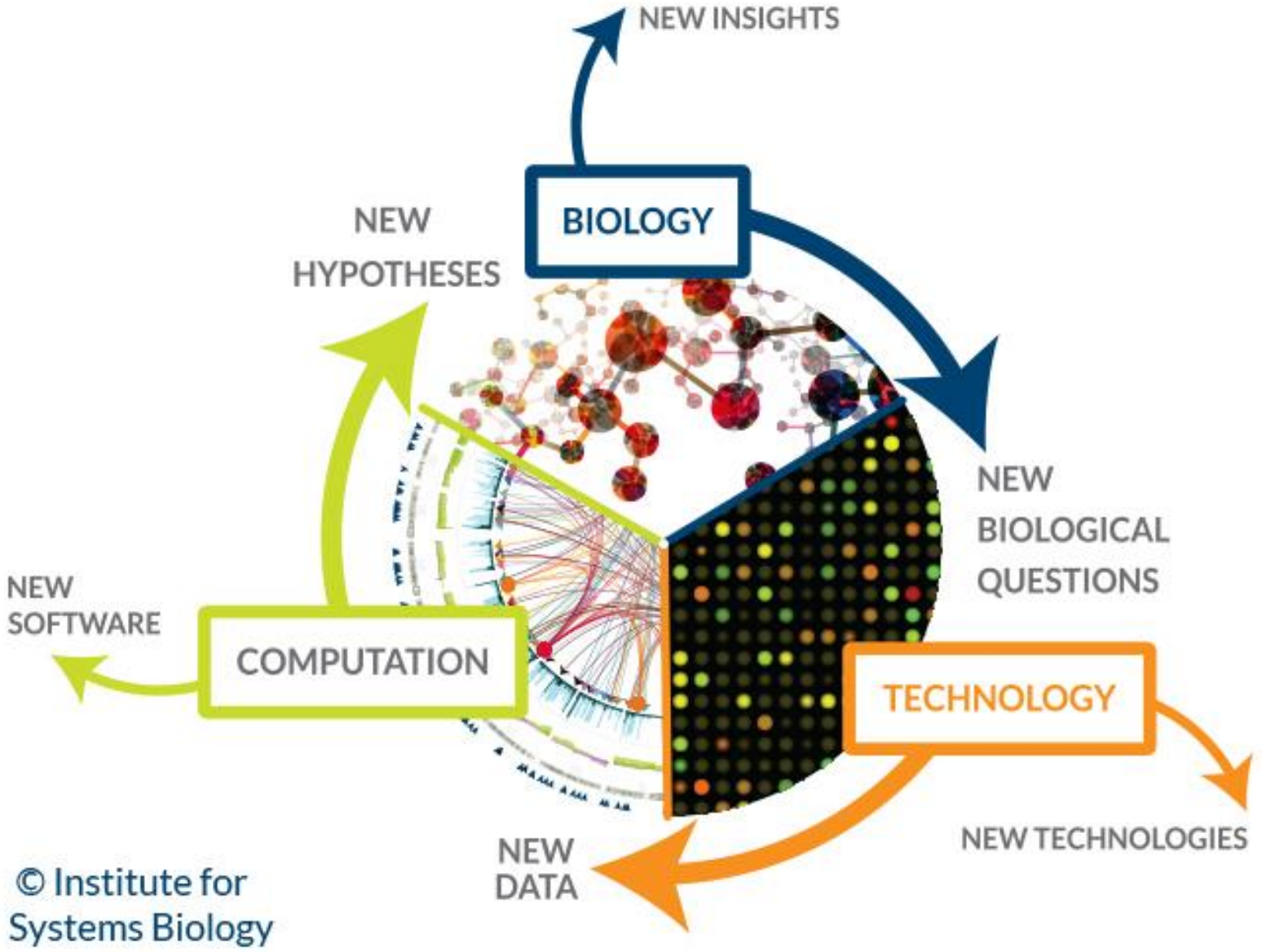
Where can I find papers?


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PubMed® comprises more than 40 million citations for biomedical literature from MEDLINE, life science journals, and online books.
Citations may include links to full text content from PubMed Central and publisher web sites.


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Case law
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Stand on the shoulders of giants

Where to Find & Evaluate Papers

- Databases: PubMed, Google Scholar, ..
- Search strategy: Combine relevant keywords, filter by recent years (5–10 yrs), select "original research article”
- Good journal criteria: Peer-reviewed, indexed, reputable publisher
- Impact Factor is not absolute: Can be inflated; a high IF does not guarantee better research
- Read critically: Evaluate whether the data truly supports the stated conclusions


NEW INSIGHTS
BIOLOGY
NEW HYPOTHESES
NEW BIOLOGICAL QUESTIONS
NEW SOFTWARE
COMPUTATION
TECHNOLOGY
NEW TECHNOLOGIES
NEW DATA
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Computational biology / Bioinformatics

- **Computational biology and bioinformatics** is an interdisciplinary field that develops and applies computational methods to analyze large collections of biological data,

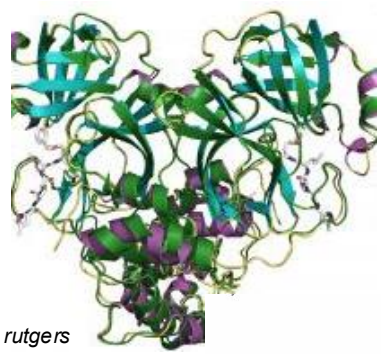
- such as genetic sequences, cell populations or protein samples, to make new predictions or discover new biology (*Nature*)



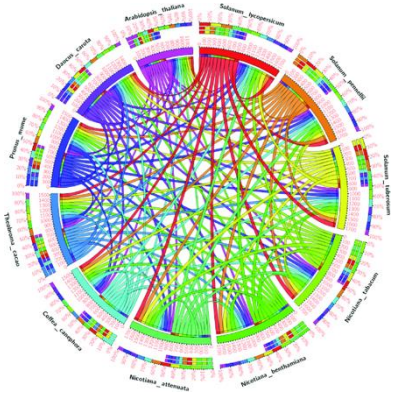
computerscience degreehub

- Bioinformatics is the application of tools of computation and analysis to the capture and interpretation of **biological data** (*NIH*)

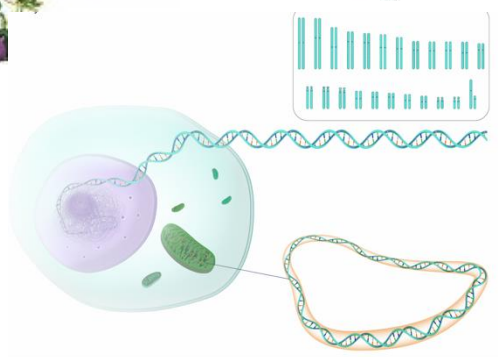
Biological Data



rutgers



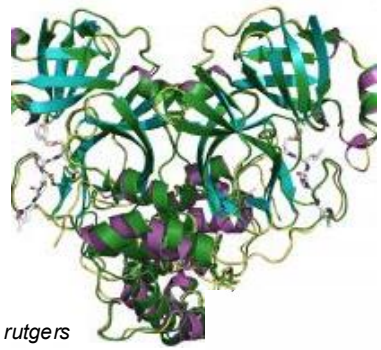
Dubey et al.



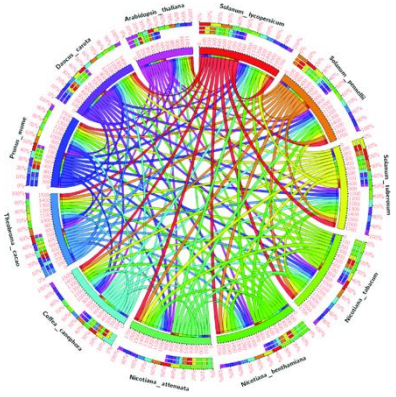
NIH

- Bioinformatics is the application of tools of **computation and analysis** to the capture and interpretation of **biological data** (*NIH*)

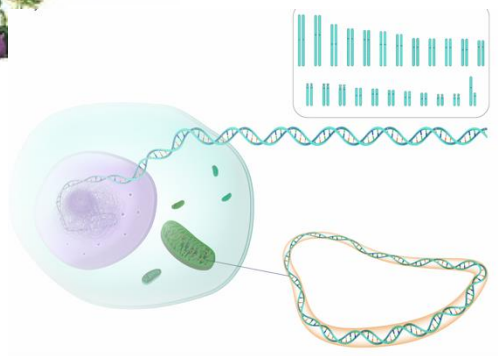
Biological Data + Computer Analysis



rutgers



Dubey et al.



NIH

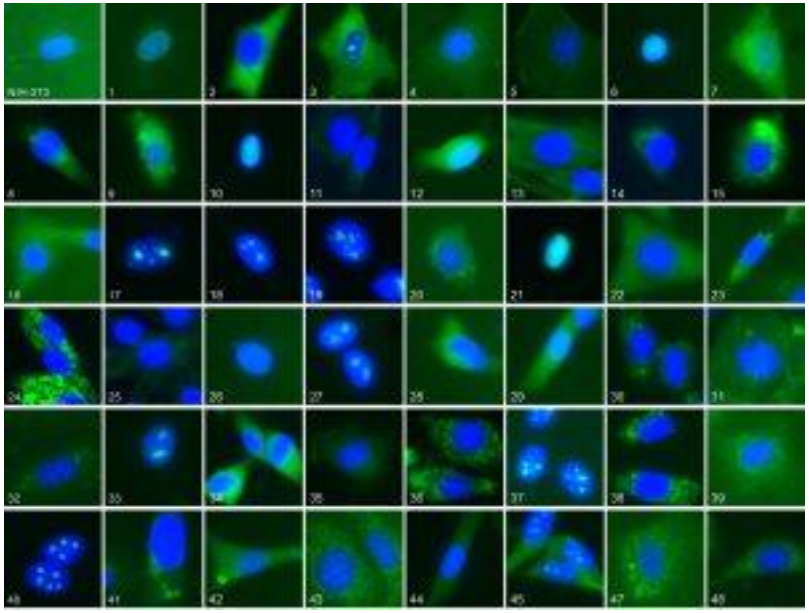


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Examples of Bioinformatics – Automated Science

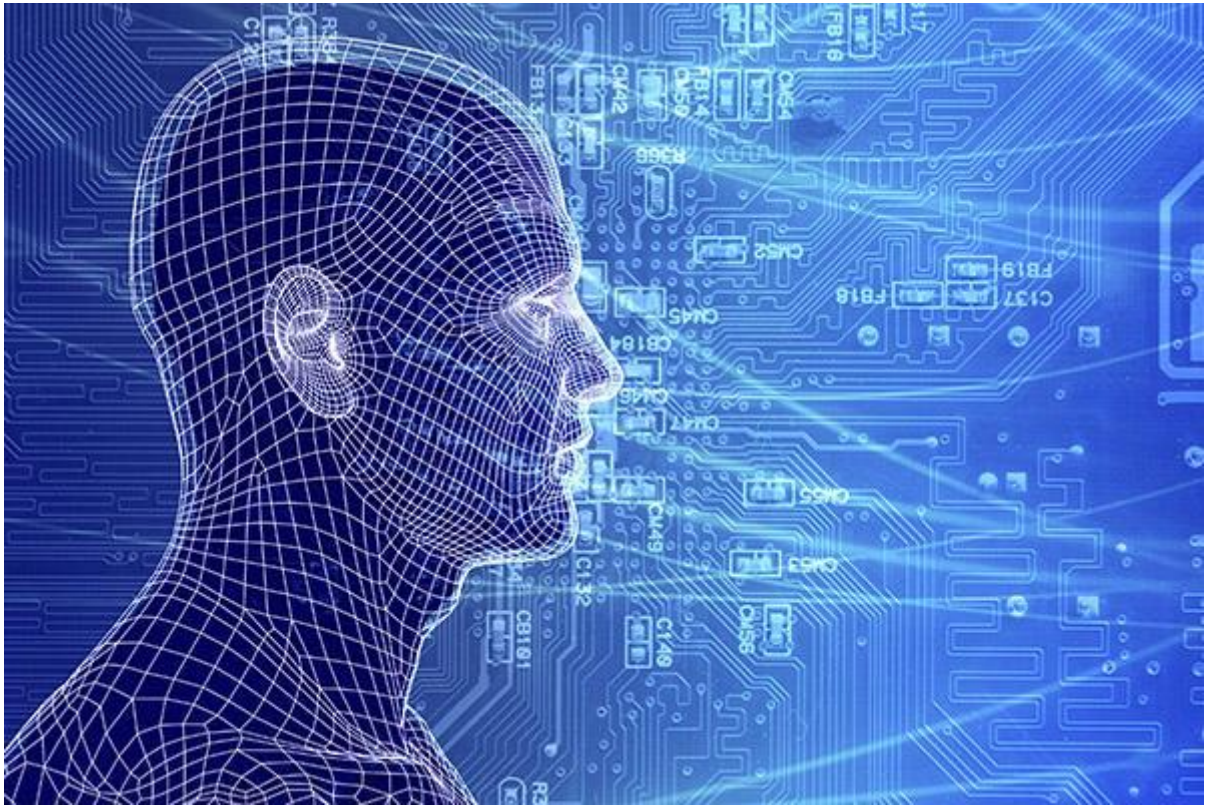


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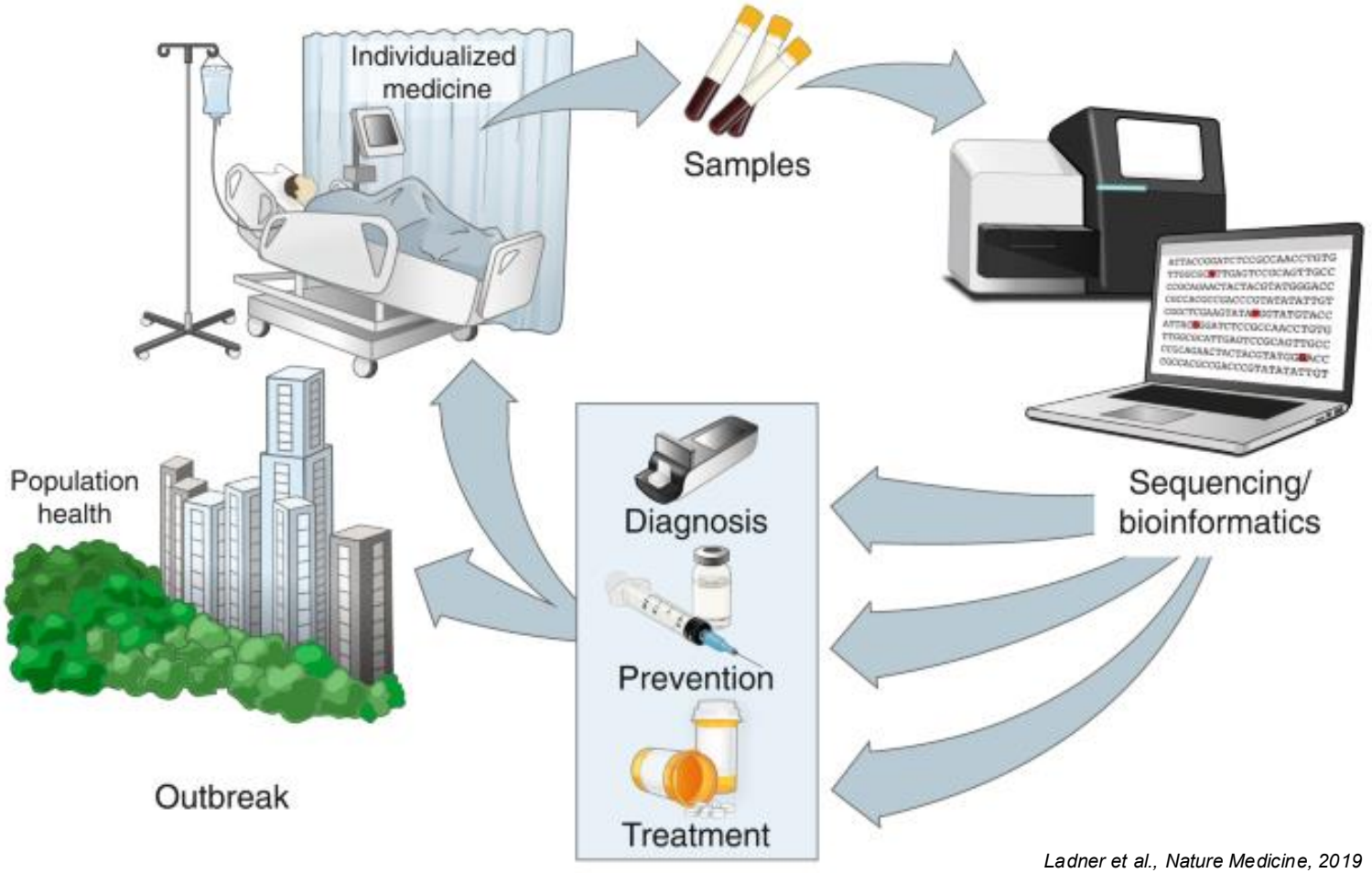
cmu

Examples of Bioinformatics – Diagnosis and Treatment of Disease



NCI

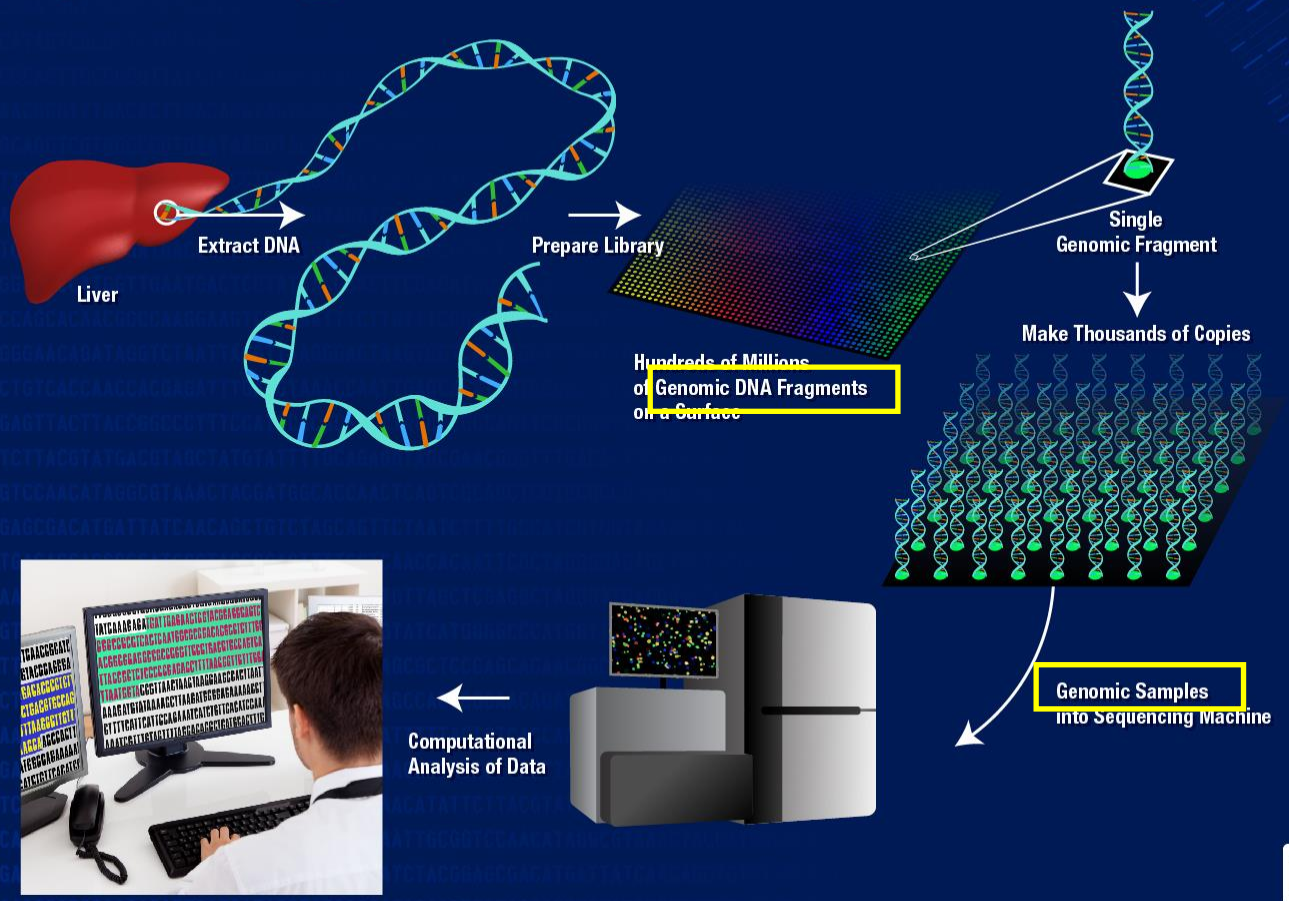
Pathogen sequencing during infectious disease outbreaks can inform precise interventions



Ladner et al., Nature Medicine, 2019

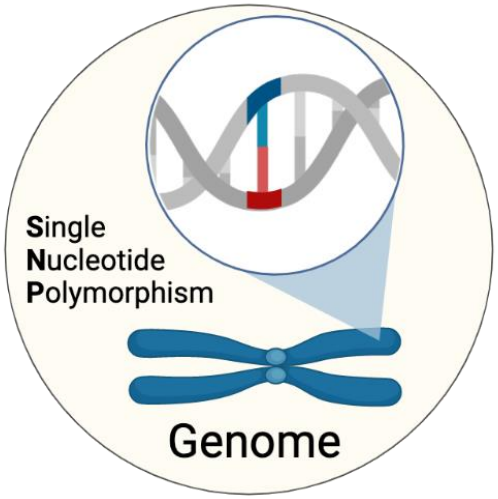
DNA Sequencing

NHGRI FACT SHEETS
genome.gov

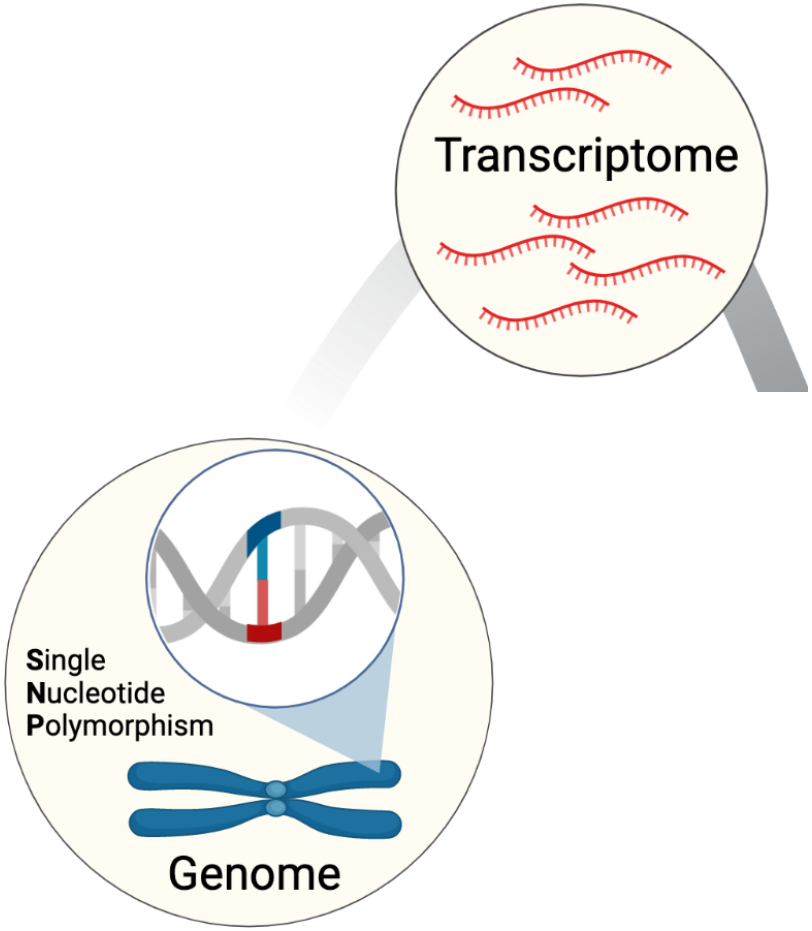


National Human Genome Research Institute

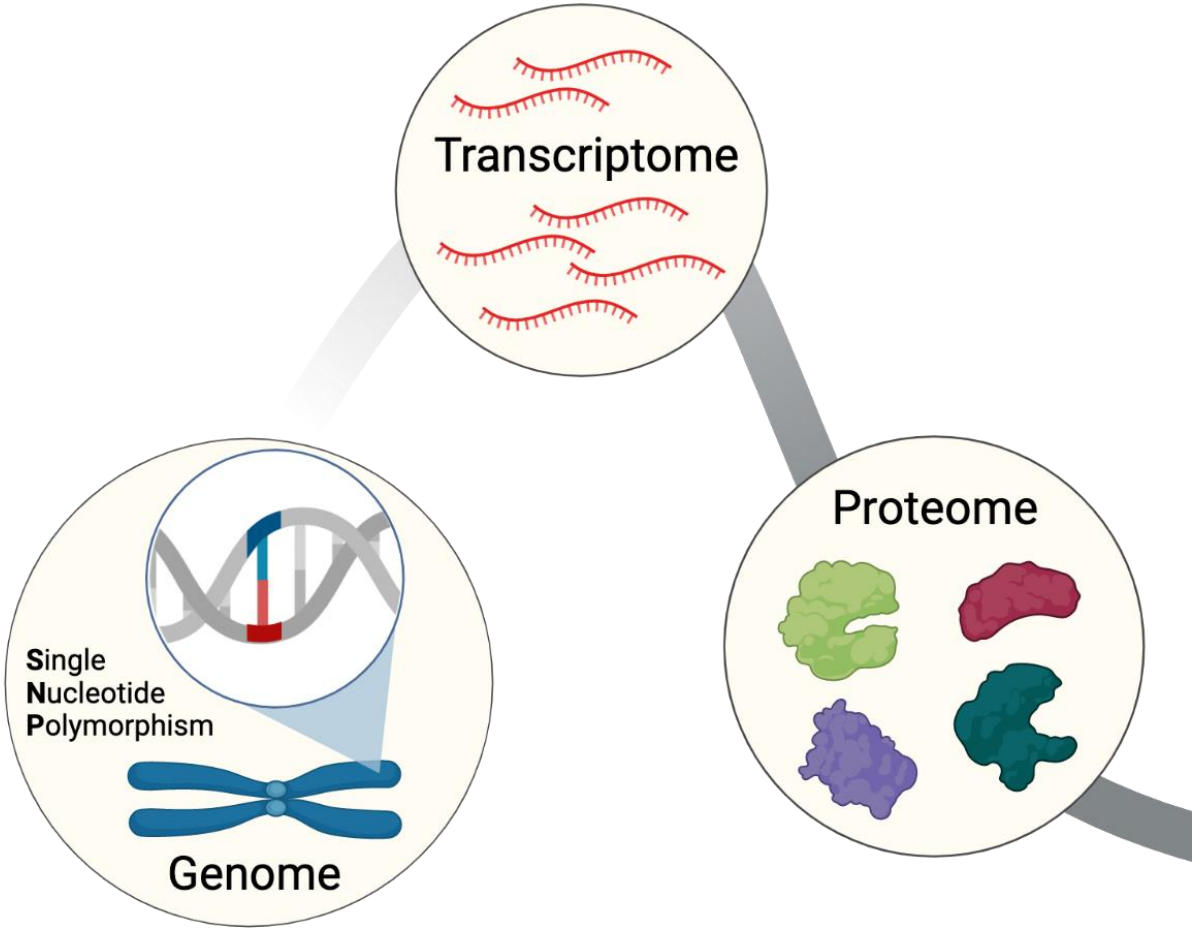
What is “genomic” data?



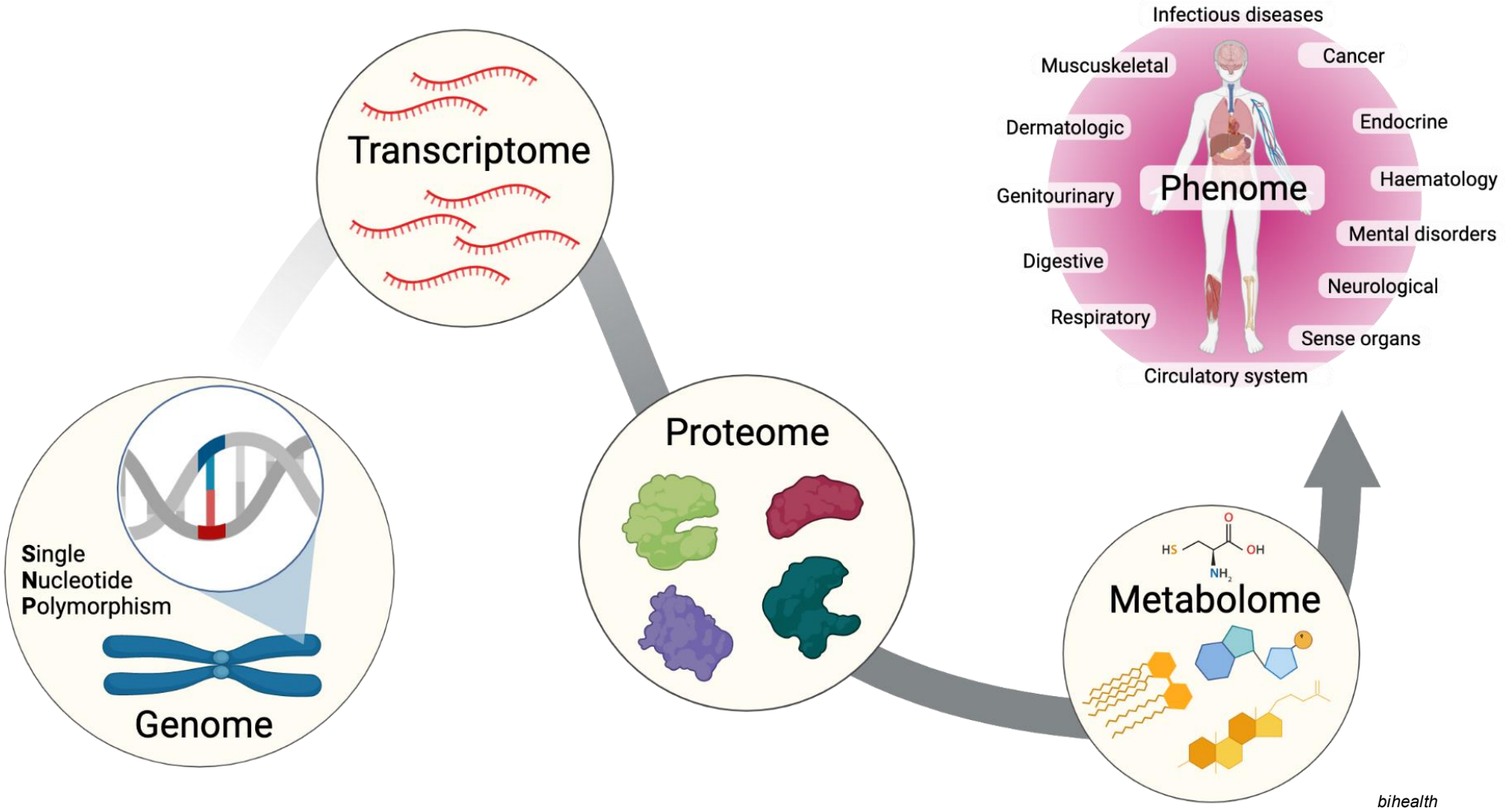
What is “genomic”/-omic data?



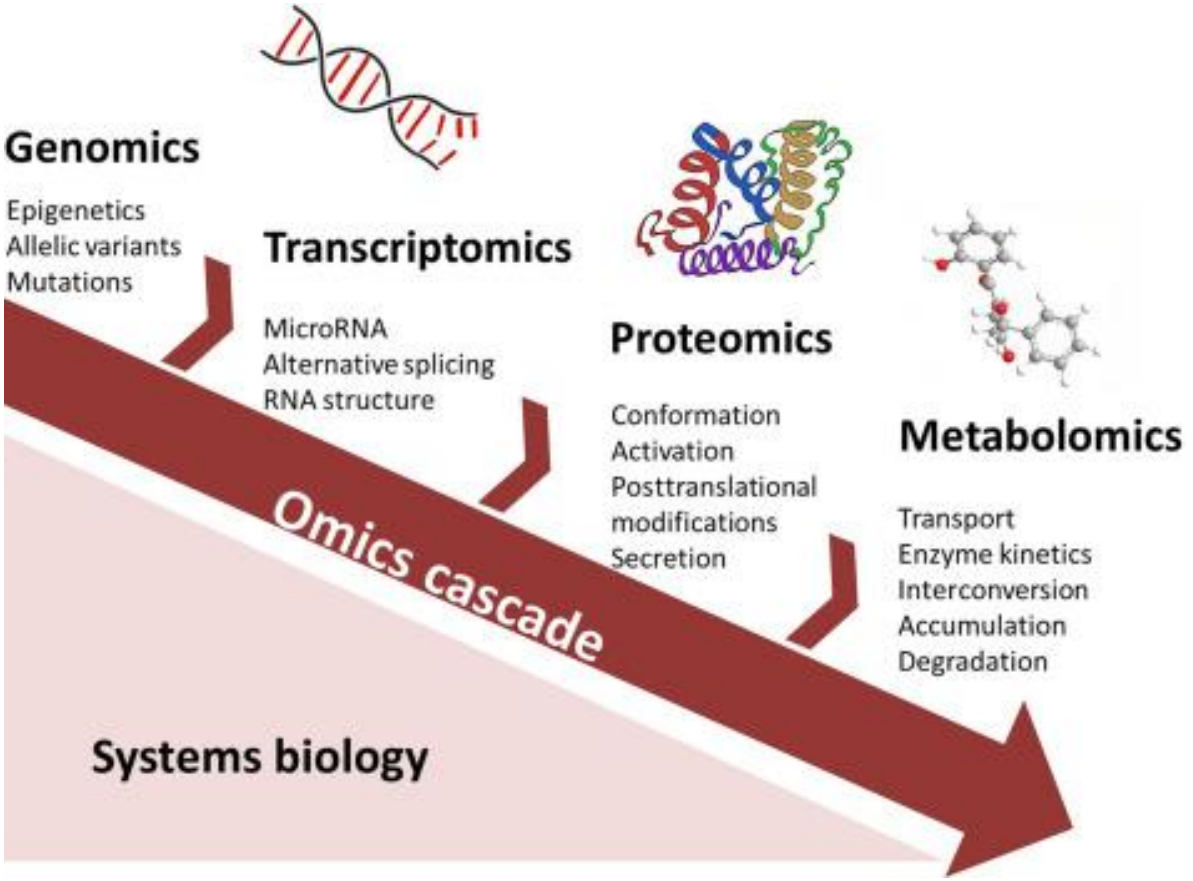
What is “genomic”/-omic data?



What is “genomic”/-omic data?

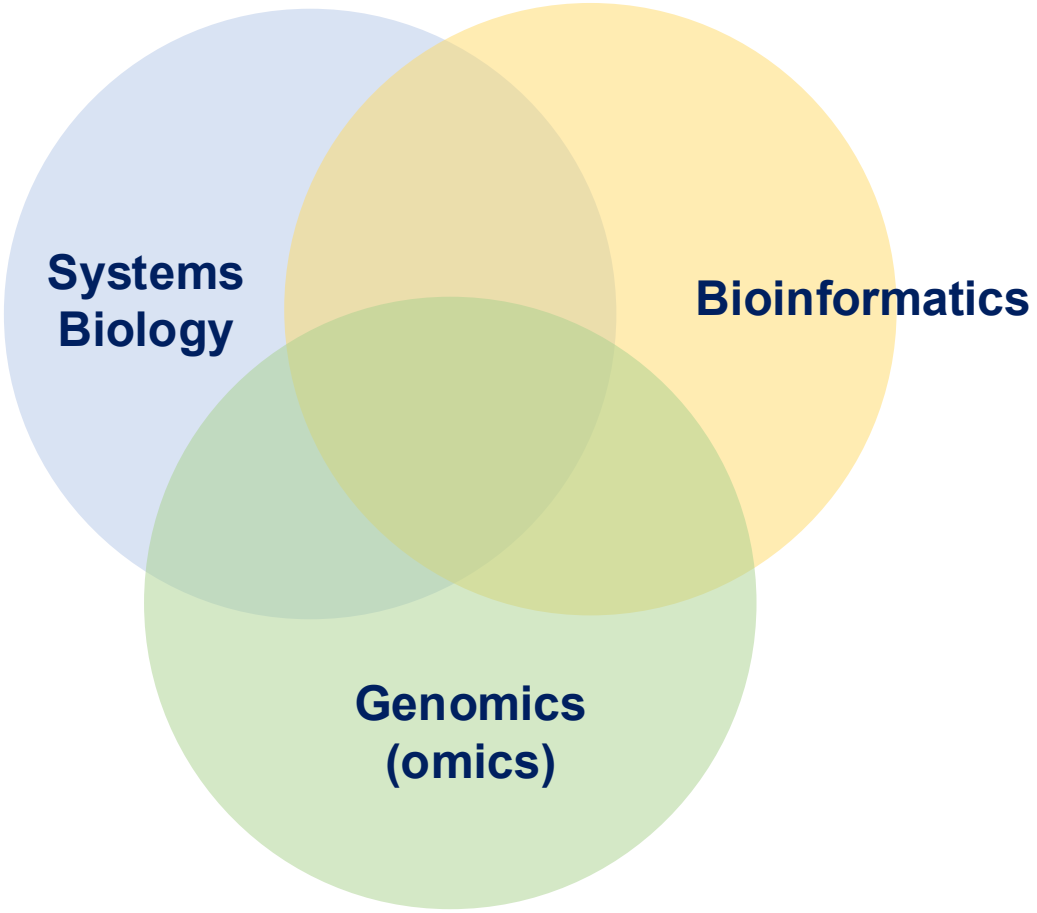


Different “omics” sciences

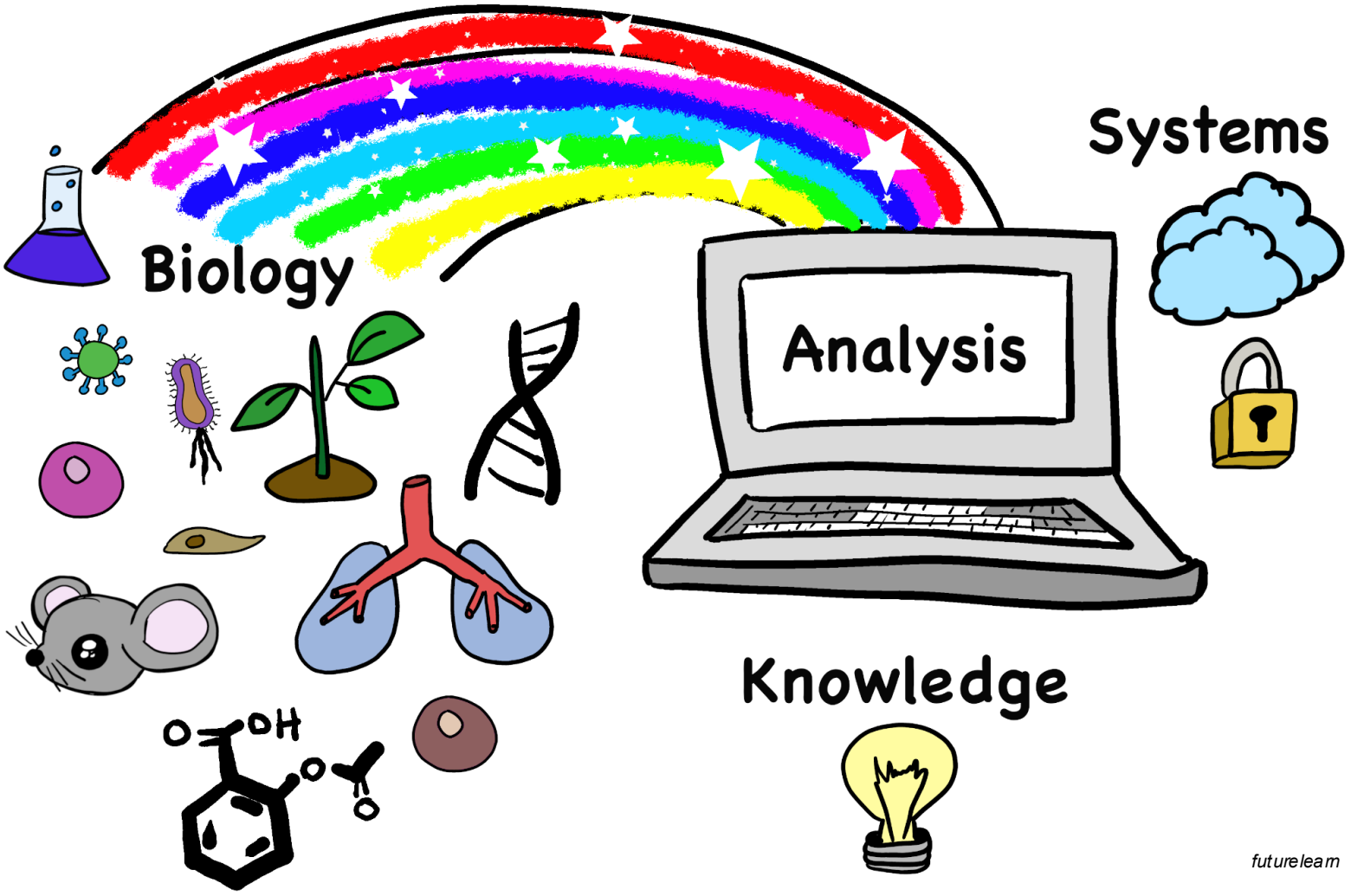


Bedia et al., 2018

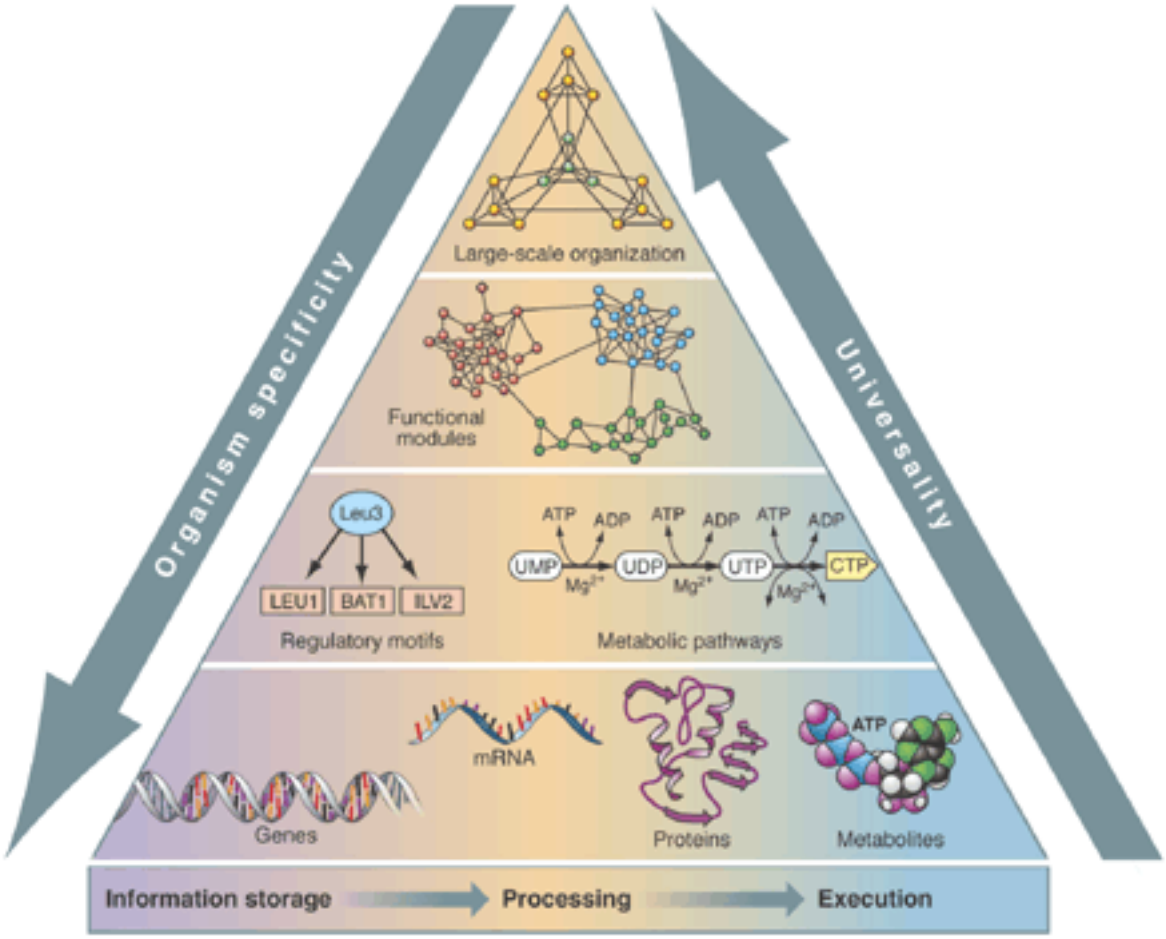
Systems Biology, Bioinformatics, and Genomics



Systems Biology, Bioinformatics, and Genomics

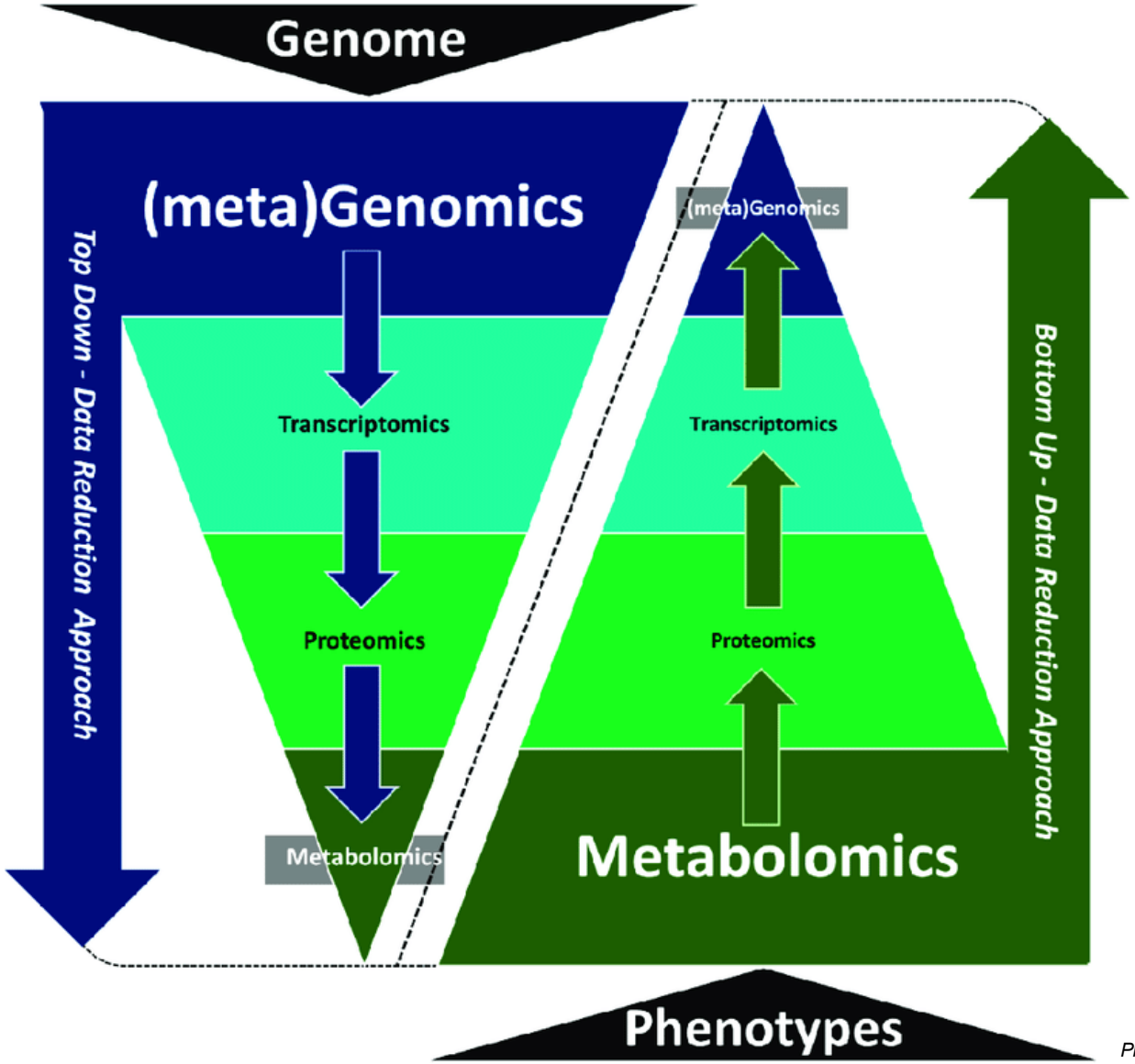


Life's Complexity Pyramid



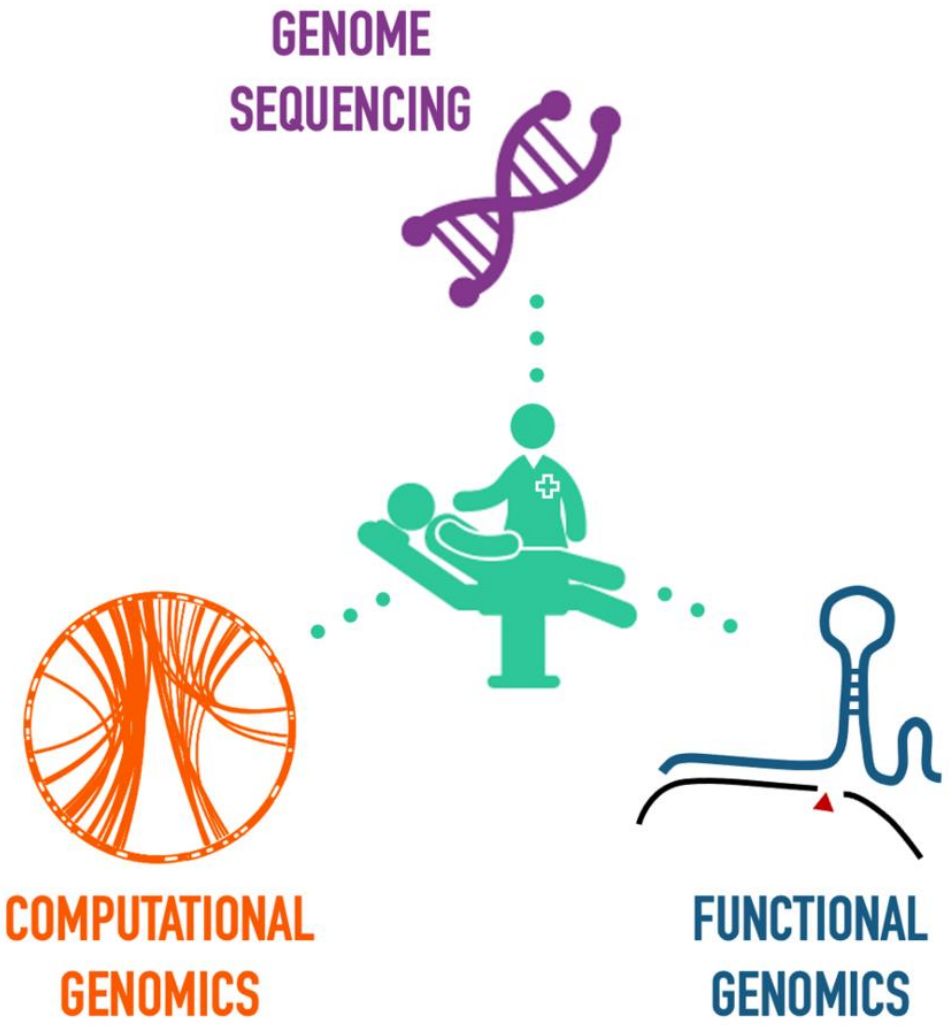
Oltvai and Barabási, Science, 2002

Data reduction integration approaches used in system biology



Pinu et al., 2019

Technologies to better understand cancer and diagnose and treat cancer patients



CCG

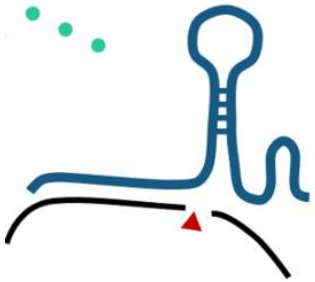
Technologies to better understand cancer and diagnose and treat cancer patients



Genome Sequencing—characterizing mutations and other molecular features of cancers

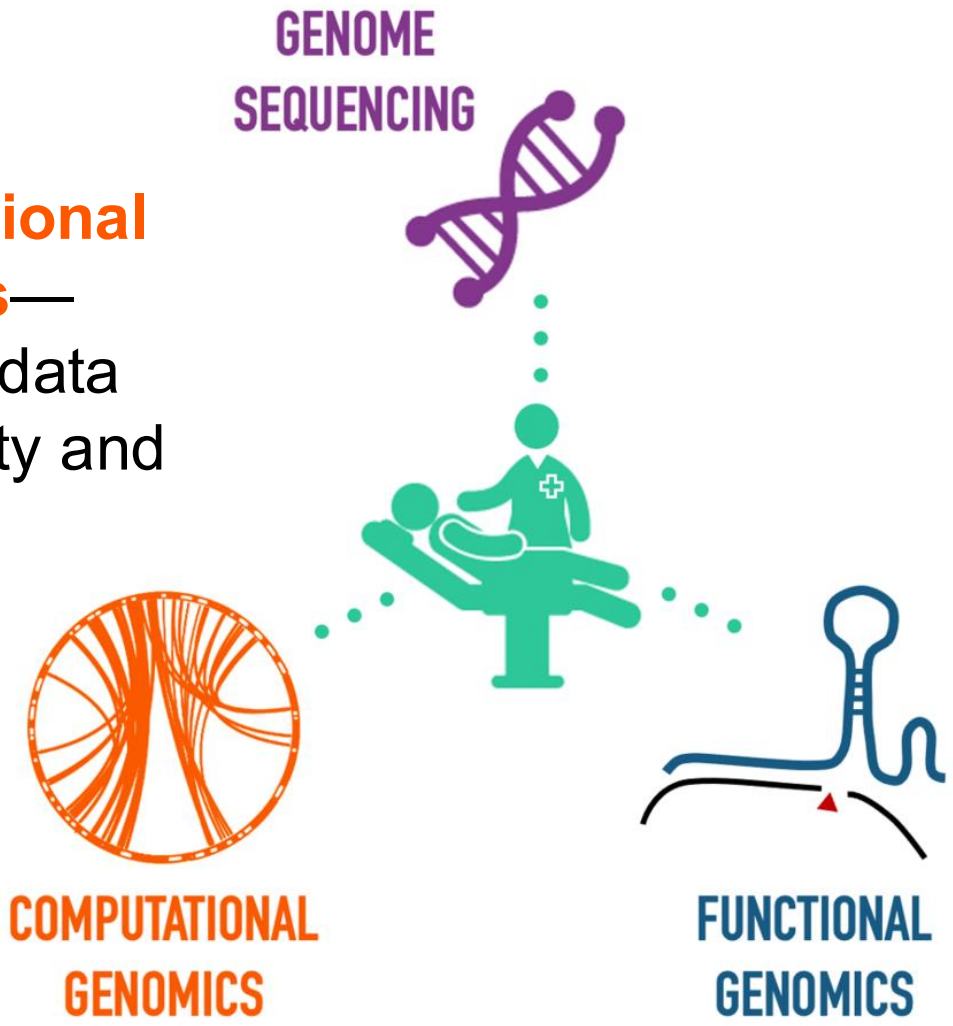
Technologies to better understand cancer and diagnose and treat cancer patients

Functional Genomics—elucidating the biological pathways underlying disease and informing precision medicine



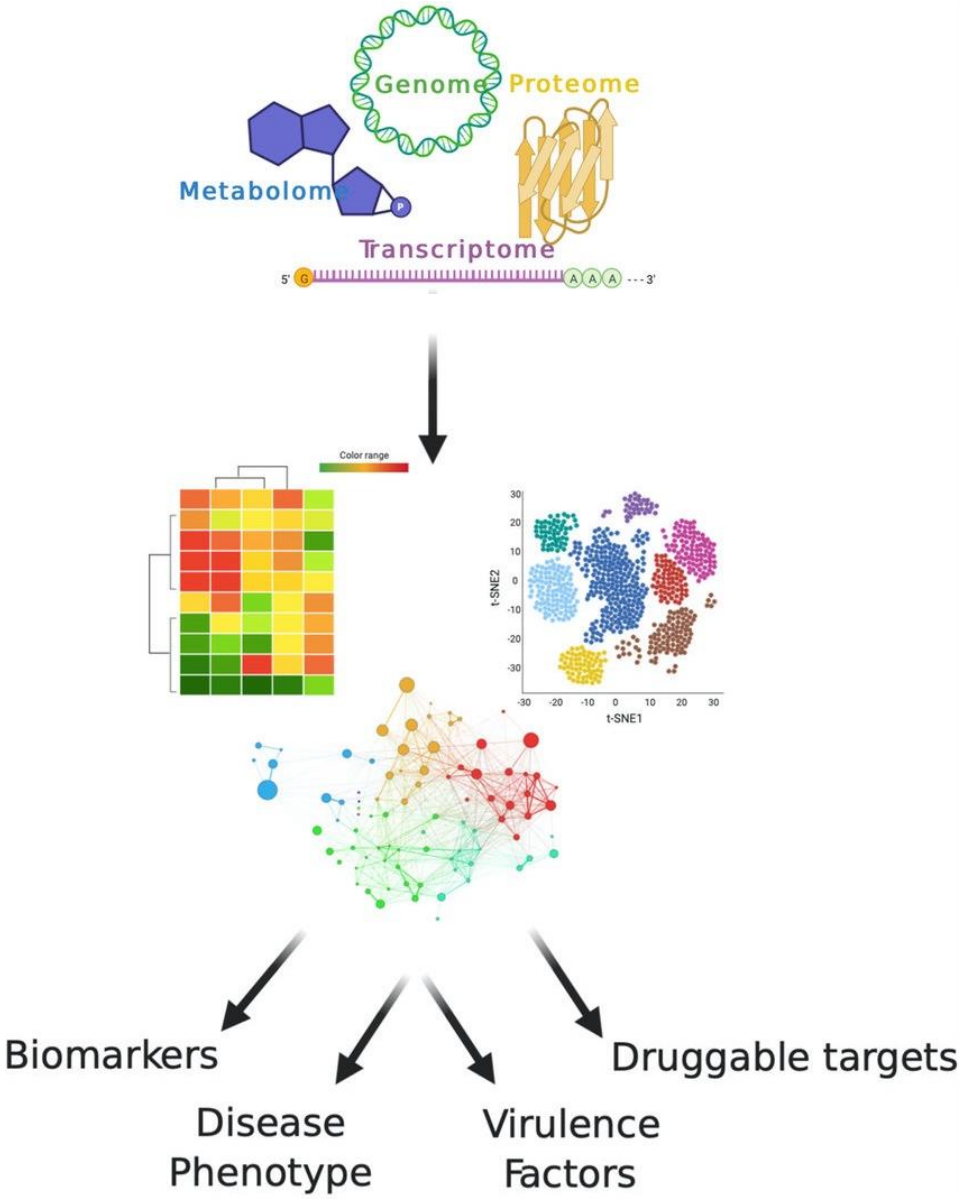
Technologies to better understand cancer and diagnose and treat cancer patients

Computational Genomics— improving data accessibility and analyses



CCG

Data Integration



Modified Ball et al., 2020

Conclusion

- **Biological systems are complex, thus, a combination of experimental and computational approaches are needed.**
- **Systems Biology can discover more complex relationships involving multiple genes.**
- **System-level studies may create new opportunities for drug discovery and better medical therapies for individual treatment.**